

Mount Kenya University

Unlocking Infinite Possibilities

DOCUMENTATION

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DECLARATION

I, **STEPHEN ONYANGO** hereby declare that all information in this document has been obtained and presented in accordance with academic rules and ethical conduct. Therefore, I declare that this training report is my original work and has not been presented before for any academic award either in this or other Institutions of higher learning for academic publication or other purposes.

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Project Title: Online Crime reporting System

Introduction

The Crime Reporting System is an online application that has been created to assist individuals to report crimes online in a secure manner. The system enables users to create, login and file crime reports describing the incident, its location, and evidence to be submitted as files. The main aim of the system is to make it easier, faster and more accessible to report crime to law enforcement agencies, to enhance citizen-to-LEA communications.

Objectives of the system

The aim of the system is to enable users to securely have accounts created, to provide the facility to users to log into the system, if they are registered users, allow victims of crime to report it online, enable evidence upload for enhanced investigations and to keep crime reports in a database for later reference. System Overview The system consists of the following major sections:

- Home Section
- User Registration
- User Login
- Crime Reporting Form
- Database Connection
- File Upload Functionality
- JavaScript Validation

Database Connection

The system is connected with MySQL database securely with PHP PDO (PHP Data Objects) technology.

Code Explanation

```
$host = "localhost";
```

```
$db_user = "root";
```

```
$db_pass = "";
```

```
$db_name = "crime_reporting_system";
```

The following parameters contain information to connect to the database. \$pdo = new PDO("mysql:host=\$host;dbname=\$db_name;charset=utf8", \$db_user, \$db_pass);

This line creates a secure link to MySQL and PHP.

\$pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION); This allows errors to be reported if there is a database error.

USER REGISTRATION MODULE

The registration section is for newcomers to register.

Features; Full name input , Email input , Password input , Password encryption

Password Security

```
$hashed_password = password_hash($password, PASSWORD_BCRYPT);
```

For security reasons, passwords are encrypted prior to being stored in the databas

Database Insertion

```
$stmt = $pdo->prepare("INSERT INTO users (fullname, email, password)
VALUES (?, ?, ?)");
```

To prevent SQL Injection attacks, prepared statements are used.

User Login Module The login module is used to enable registered users to logon to the system. **Login Process** The system checks:

- **Whether the email is present or not.**
- whether the password matches the encrypted password stored in the database.

Password Verification

```
password_verify($password, $user['password'])
```

This function compares the entered password with the encrypted password stored in the database.

Crime Reporting Module

Users can report incident information on the report form.

Information Collected includes, Incident title, Description, Location,

Evidence file

File Upload

```
move_uploaded_file($temp_name, "uploads/" . $evidence); Saves evidence
file(s) into uploads folder.
```

Security Feature

```
$evidence = time() . "_" . basename($evidence);
```

The uploaded files are stamped with a date so as not to replace each other.

Frontend Design

HTML and CSS were used to develop the frontend.

Features

- **Navigation bar**
- **Responsive forms**
- **Hidden sections**
- **Interactive buttons**

Styling

CSS was used to Add colors, Create spacing, Style forms Add shadows and borders to the flower.

Javascript Validation

To make sure the user fills out all of the required fields before submitting, JavaScript validation is employed. Example `if(password.length < 6){ alert ("The password is too short, it should be at least 6 characters long"); }` This makes the user experience better and decreases the amount of invalid submissions.

Section Navigation

The web site dynamically displays and hides sections with JavaScript.

Function Used `showSection(section)` This function controls the Home page, Login page, Registration page, Report page

Database Table

User Table

Field	Type
id	INT
fullname	VARCHAR
email	VARCHAR
password	VARCHAR

Report Table

Field	Type
id	INT
title	VARCHAR
description	TEXT
location	VARCHAR
evidence	VARCHAR

Registration Workflow

System Workflow

1. User opens registration form.

2. User enters details.
3. Password is encrypted.
4. Data is saved in database.

Login Workflow

1. User enters email and password.
2. System verifies credentials.
3. User accesses system.

Crime Reporting Workflow

1. User fills report form.
2. User uploads evidence.
3. System stores report in database.
4. Confirmation message appears.

Security Features

The system includes several security measures:

- Password hashing
- Prepared SQL statements
- File upload handling
- Input validation

These features help protect user data and prevent attacks.

Challenges faced

The following challenges were faced while developing; Database connection errors, File upload configuration, Form validation issues, Receiving multiple email addresses as registrants. The debugging and testing helped to overcome these challenges.

Testing

The system was tested such that; Registration works correctly, Logging in is working correctly, Reports have been saved successfully, Evidence uploads work and a stable connection to the database.

Conclusion

The Crime Reporting System is an effective tool for citizens to safely and effectively submit reports electronically. The project is used to showcase PHP, MySQL, JavaScript, HTML and CSS in creating a working web-based application. The system makes something more accessible and digital method of crime reporting.

Future Improvement

Future enhancements to be considered include:

1. Admin dashboard
2. Email notifications
3. GPS location tracking
4. Live crime updates
5. Mobile application integration
6. Report tracking system